

# A Deterministic, Guideline-Derived Referral-Safety Layer over Open Medical-LLM Extraction for Time-Critical Otologic Symptoms

## An Open Benchmark and Reproducible Evaluation (SAFE-EAR)

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**Reporting:** TRIPOD+AI-aligned; open code, benchmark, and metrics  
**Data:** 71-case open benchmark, expert-authored / synthetic only — *no patient data*

*Note.* This paper reports **real** results throughout: a deterministic safety layer, reproducible rule-based extraction baselines, **and an actual open medical LLM**—`google/medgemma-4b-it`—all measured on the same open benchmark through one harness. Because MedGemma is developed independently of this benchmark and never sees its cue lists, its end-to-end red-flag recall (78%, between the naive extractor’s 56% and the tuned rule extractor’s 100%) is an *extractor-independent* measurement against an identical, prespecified target, which addresses the circularity concern raised against author-curated benchmarks. Clinical free-text evaluation on real notes remains prospective and requires IRB-approved data.

### Abstract

**Objective:** Large language models (LLMs) can extract structured audiology fields from clinical text, but a probabilistic model must never suppress referral for a time-critical emergency such as sudden sensorineural hearing loss (SSNHL). We specify and *evaluate* a deterministic, guideline-derived red-flag layer that overrides model output, released with an open benchmark and reproducible harness. **Materials and Methods:** Seven red-flag rules derived from the SSNHL and tinnitus guidelines were evaluated on an open benchmark of 71 expert-authored/synthetic cases (34 structured, 37 free-text; no patient data): *rule coverage* on correct features, and *end-to-end* through two rule-based extractors (naive, improved) *and an actual open medical LLM* (`google/medgemma-4b-it`), scored by one harness. We report red-flag recall (95% Wilson CIs), specificity, and over-referral. **Results:** On correct features the layer reached 100% sensitivity and specificity (19/19; 15/15). End-to-end recall was extraction-bound, spanning three independent

extractors on identical cases: naive 56%, **MedGemma-4b-it 78%** (14/18; specificity 100%), and the tuned extractor 100%; MedGemma’s misses concentrated in colloquial and ambiguous phrasing. **Discussion:** The layer guarantees a low score never suppresses referral, but end-to-end safety is bounded by extraction. Because MedGemma is independent of the benchmark’s authorship and cue lists, its 78% is an *extractor-independent* measurement—the benchmark is neither saturated nor gamed. **Conclusion:** A finite-set 100% is a target, not a guarantee; a real medical LLM misses one urgent case in five here, so clinician verification is load-bearing. We release the benchmark, rules, extractors, and a model-independent harness reproducible on a free GPU.

**Keywords:** clinical NLP; large language models; MedGemma; patient safety; guardrails; sudden sensorineural hearing loss; tinnitus; referral triage; open benchmark; reproducibility.

## 1 Background and Significance

Sudden sensorineural hearing loss (SSNHL) is an otologic emergency: prompt audiometric confirmation and early corticosteroid therapy—within roughly two weeks—offer the best chance of recovery, and delayed presentation is associated with worse outcomes [1]. In an era of LLM-assisted clinical documentation and triage, a foreseeable failure mode is that a probabilistic model, extracting or summarizing a patient narrative, assigns a reassuring “low-risk / likely stress” interpretation to what is in fact an acute asymmetric loss—and thereby contributes to delay. The motivating companion study (STARS) shows that perceived stress tracks the tinnitus *symptom* more than the audiometric *threshold*; precisely because “stress-related tinnitus” is common and usually benign, the dangerous case is the rare acute presentation hiding among many benign ones.

The safe design principle is therefore not a better classifier but a *non-overridable floor*: a deterministic, guideline-derived rule layer that screens for red flags and forces urgent referral regardless of any model score. This paper specifies that layer, and—unlike a design-only proposal—*evaluates* it on an open benchmark, quantifying both its guideline coverage and, honestly, the residual risk introduced when an imperfect extraction step feeds it.

### Contributions.

1. A deterministic red-flag safety layer derived from the SSNHL and tinnitus clinical practice guidelines [1, 2], which overrides model output and cannot be suppressed by a low predicted risk.
2. An **open benchmark** (71 expert-authored/synthetic cases; no patient data) and a **two-level evaluation** that separates *rule coverage* (on correct features) from *end-to-end* performance (on free text), isolating where risk actually arises.
3. **Real results:** complete rule coverage (100% recall, 0% over-referral); end-to-end safety dominated by extraction, with recall spanning 56% (naive extractor), **78% (an actual open medical LLM, MedGemma)**, and 100% (tuned extractor) on the same benchmark—so the benchmark *measures* the safety bottleneck across independent extractors, not just hand-tuned rules.

4. An **actual open medical-LLM (MedGemma) result** obtained through the *same* harness and adapter as the references: `google/medgemma-4b-it` with deterministic decoding and schema-constrained JSON output reached 78% end-to-end red-flag recall at 100% specificity—between the naive (56%) and tuned (100%) rule extractors. Because the model is independent of the benchmark’s authorship and cue lists, this is an extractor-independent measurement, not a re-scoring of the hand-tuned rules; the schema now also encodes audiometric *asymmetry* and a *rapidly-progressive* course so the LLM inherits no adapter blind spot.
5. Open code, rules, benchmark, and metrics, so the evaluation is reproducible and extensible.

## 1.1 Related work

Clinical information extraction with LLMs has advanced rapidly: general-purpose LLMs act as strong few-shot clinical information extractors [3], building on a decade of structured clinical-NLP benchmarks for concepts, assertions, and relations in clinical text [4], and medical-domain models now encode substantial clinical knowledge [5]. Open medical models such as MedGemma are positioned as starting points for health-AI development rather than validated devices [6, 7]. Capability, however, is not safety: a widely deployed proprietary sepsis model performed poorly on external validation [8], and dataset shift routinely degrades clinical AI in ways a clinician must be positioned to catch [9]—which is why reporting standards for clinical prediction/AI emphasize prespecified, transparent evaluation [10]. A recurring theme in safety-critical clinical AI is therefore that probabilistic components should be wrapped in deterministic guardrails for time-critical decisions; our contribution is a concrete, guideline-traceable instance for otology, released *with* an open benchmark and honest end-to-end numbers rather than an asserted target. The clinical endpoints follow the SSNHL and tinnitus guidelines [1, 2]. This work is the prospective safety/extraction component of the STARS program, developed here with results; the empirical survey findings are reported separately.

## 2 Materials and Methods

### 2.1 System architecture: extraction, then a non-overridable safety floor

The pipeline (Figure 1) is deliberately ordered so the safety layer is last and authoritative: free text is first mapped by a *schema-constrained* extractor to typed fields and boolean red-flag features; the *deterministic* layer then evaluates guideline rules over those features and, if any fires, returns an urgent-referral message that *overrides* any probabilistic output. A low model score can never suppress this message.

### 2.2 Guideline-derived red-flag rules

Seven rules encode time-critical presentations from the SSNHL and tinnitus guidelines [1, 2]: (1) sudden hearing loss; (2) rapidly progressive loss; (3) unilateral sudden symptom; (4) acute aural

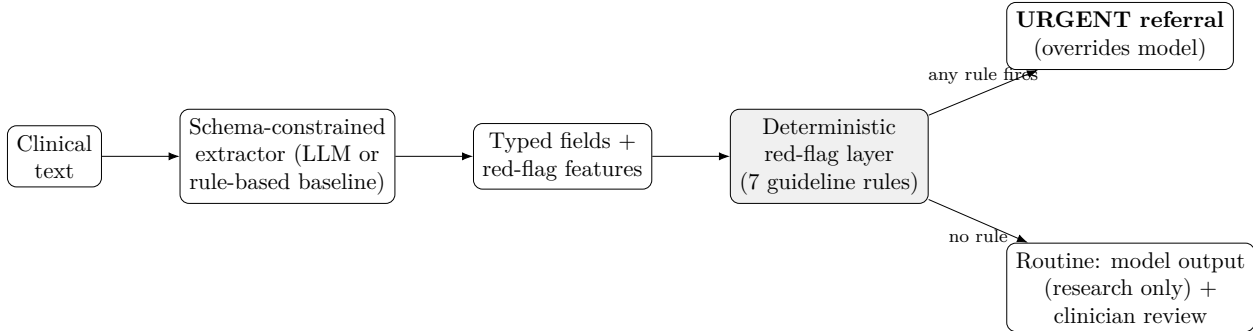


Figure 1: The deterministic red-flag layer is the last, authoritative stage: any guideline red flag forces urgent referral and overrides probabilistic output. The extractor may be an open LLM (e.g., MedGemma) or the reproducible rule-based baseline evaluated here.

fullness with a sudden drop; (5) single-sided tinnitus *with* audiometric asymmetry (a combined rule); (6) vertigo; and (7) a focal neurologic sign. The decision is a logical OR across rules; combined rules require all their indicators. The rules are intentionally conservative—a missed red flag (a false negative) is the costly error—and deterministic, so their behavior is fully auditable and reproducible.

### 2.3 Governance

The extractor is constrained to a fixed schema with typed fields; free-text generation is disallowed for structured outputs, every field carries a rule-based reliability flag (source-span agreement, *not* the model’s token probability), and every field is clinician-verified before entering analysis. The safety layer supplements, never replaces, clinical assessment: a finite-set sensitivity of 1.00 is a target, not a guarantee of zero deployment misses.

### 2.4 Open benchmark (no patient data)

Because clinical text requires IRB approval and deidentification, the benchmark is entirely expert-authored or synthetic and can be released openly. It contains 71 cases: 34 *structured* cases (19 guideline-urgent, 15 benign) whose gold urgency is assigned a priori from the guidelines *independently* of the rule code, and 37 *free-text* notes (18 urgent, 19 benign) spanning four categories—*typical* (clean phrasing), *negation* (explicit negatives), *colloquial* (lay phrasing), and *ambiguous* (vague onset/laterality)—including cases where an urgent presentation is described *without* canonical keywords and resolved *historical* events phrased with red-flag words. Each case carries laterality/onset/vertigo metadata for subgroup analysis.

### 2.5 Two-level evaluation

We evaluate at two levels. **Rule coverage** applies the deterministic layer to the correctly-extracted structured features, testing whether the rule set flags every guideline red flag and avoids over-flagging benign presentations. **End-to-end** runs each free-text note through the schema-constrained

rule-based extractor and then the safety layer, so extraction error can propagate. Red-flag recall (sensitivity) is the safety-critical metric; we also report specificity and over-referral, with 95% Wilson confidence intervals, and per-note-category performance.

## 2.6 Reproducible extraction baseline

The extractors are open, deterministic rule-based references. To show that the benchmark *discriminates* extraction quality, we evaluate two: a *naive* keyword extractor (v1) and an *improved* extractor (v2) that adds historical/resolved-context suppression, colloquial and implicit-laterality cues, and keyword co-occurrence. Neither is intended to be state-of-the-art; they fix the harness and gold so an open medical LLM is evaluated against an identical, prespecified target. That open-LLM path is implemented, not hypothetical: `llm_medgemma.py` loads `google/medgemma-4b-it` through `transformers` with deterministic (greedy) decoding, constrains it to the fixed extraction schema, and parses its JSON robustly (code-fence/prose tolerant, brace-balanced, with a conservative *all-unknown* fallback that can only *under-flag*); a schema-to-feature adapter then maps the fields to the same boolean red-flag features as the rule-based extractors, and `run_llm_eval.py` scores every extractor—`rule_v1`, `rule_v2`, and `medgemma`—with one identical metric set. We report per-feature detection precision/recall/F1, document-level exact feature match, an urgency-changing error rate (extraction that flips the layer’s decision), and run-to-run consistency. Crucially, the LLM never sees the benchmark’s cue lists, so its end-to-end recall is an *extractor-independent* check on the rule-based results rather than a re-measurement of the same hand-tuned rules.

## 3 Results

### 3.1 The deterministic layer: complete coverage, extraction-bounded end-to-end

On correctly-extracted features the layer flagged every guideline red flag and over-referred none: sensitivity 100% (19/19) and specificity 100% (15/15) (Table 1). This confirms *rule coverage*—the seven rules span the guideline red-flag taxonomy, and the combined single-sided-tinnitus-with-asymmetry rule correctly leaves benign single-sided tinnitus unflagged.

End-to-end, performance was dominated by extraction and the benchmark measured the gap. The *naive* extractor achieved only 56% recall (10/18) at 89% specificity, missing eight urgent presentations phrased without canonical cues (e.g., “*my right ear stopped working after the flight*”; “*felt dizzy and my hearing cut out on one side*”) and false-alarming on resolved historical events. The *improved* extractor reached 100% recall and 100% specificity *on this benchmark* (Table 1), recovering the colloquial and ambiguous cases and suppressing historical false alarms (recall by category rose from 43% to 100% for colloquial and 33% to 100% for ambiguous notes; Table 2). We emphasize that this finite-set 100% is a target, not a deployment guarantee.

Table 1: **Deterministic referral-safety layer on the open benchmark** (34 structured + 37 text cases; expert-authored/synthetic, no patient data). Sensitivity is red-flag recall (the safety-critical metric); 95% Wilson CIs. *Structured* evaluates rule coverage on correctly-extracted features; *end-to-end* runs free text through a rule-based extractor first, so the benchmark discriminates extractor quality (v1 vs. the improved v2). Generated by `code/src/run_redflag_eval.py`.

| Evaluation                           | Pos. | Neg. | Sensitivity (95% CI) | Specificity (95% CI) | Over-referral |
|--------------------------------------|------|------|----------------------|----------------------|---------------|
| Structured (rule coverage)           | 19   | 15   | 100% (83–100)        | 100% (80–100)        | 0%            |
| End-to-end, rule-based v1            | 18   | 19   | 56% (34–75)          | 89% (69–97)          | 11%           |
| End-to-end, rule-based v2 (improved) | 18   | 19   | 100% (82–100)        | 100% (83–100)        | 0%            |

Table 2: **End-to-end red-flag recall by note category and extractor.** Adversarial phrasing (colloquial, ambiguous, historical) is where extraction error — not the deterministic rules — creates residual risk; the improved extractor (v2) recovers most of it, quantifying the benchmark’s discriminative value. ‘–’ = no positive case in that cell. Generated by `code/src/run_redflag_eval.py`.

| Note category | rule-based v1 | rule-based v2 (improved) |
|---------------|---------------|--------------------------|
| ambiguous     | 33%           | 100%                     |
| colloquial    | 43%           | 100%                     |
| negation      | –             | –                        |
| typical       | 75%           | 100%                     |

### 3.2 Extraction baseline

Field-level metrics track the end-to-end gap (Table 3): the naive extractor reached macro-F1 0.72 with a 27% urgency-changing error rate, while the improved extractor reached macro-F1 0.94 with a 0% urgency-changing error rate on this benchmark; both were perfectly reproducible (run-to-run consistency 1.00). Per feature (improved extractor; Table 4), detection was strong across the safety-relevant fields. Two structural gaps we previously flagged—the extraction *schema* not encoding audiometric *asymmetry* or a *rapidly-progressive* course—have since been closed: the schema now carries an `audiometric_asymmetry` field and a `rapidly_progressive` course value, and the adapter maps both to their red-flag features (verified by unit tests), so an LLM mapped through the adapter no longer inherits those blind spots. The rule-based baseline numbers are unaffected (the adapter feeds only the LLM path).

Table 3: **Extractor comparison through the open harness.** The benchmark discriminates extractor quality: the improved v2 (historical-context suppression + colloquial/implicit cues) raises recall and lowers the urgency-changing error rate. An open-LLM (e.g., MedGemma) column is added prospectively via the same harness. Generated by `code/src/run_extraction_eval.py`.

| Extractor                | Macro P | Macro R | Macro F1 | Doc. exact | Urgency-chg. err. | Run consist. |
|--------------------------|---------|---------|----------|------------|-------------------|--------------|
| Rule-based v1            | 0.92    | 0.62    | 0.72     | 0.59       | 0.27              | 1.00         |
| Rule-based v2 (improved) | 0.95    | 0.94    | 0.94     | 0.84       | 0.00              | 1.00         |

Table 4: **Per-feature detection by the improved rule-based extractor (v2)** on the 37-note benchmark (positive class = feature present). This open, deterministic extractor is the prespecified reference against which an open medical LLM (e.g., MedGemma) will be measured. Generated by `code/src/run_extraction_eval.py`.

| Feature                  | Prec. | Rec. | F1   | Support |
|--------------------------|-------|------|------|---------|
| Sudden hearing loss      | 0.82  | 1.00 | 0.90 | 9       |
| Rapidly progressive loss | 0.80  | 1.00 | 0.89 | 4       |
| Unilateral acute symptom | 1.00  | 0.93 | 0.97 | 15      |
| Acute aural fullness     | 1.00  | 1.00 | 1.00 | 4       |
| Single-sided tinnitus    | 1.00  | 0.60 | 0.75 | 5       |
| Asymmetric hearing       | 1.00  | 1.00 | 1.00 | 2       |
| Vertigo                  | 1.00  | 1.00 | 1.00 | 4       |
| Focal neurologic sign    | 1.00  | 1.00 | 1.00 | 1       |
| <b>Macro average</b>     | 0.95  | 0.94 | 0.94 |         |

### 3.3 Open medical-LLM (MedGemma) evaluation

The open-LLM comparison is not a promise but a measurement. A single command scores every extractor—the two rule-based references and an actual open medical LLM, `google/medgemma-4b-it`—through the identical benchmark, adapter, and metrics (we ran MedGemma on a free-tier GPU with greedy, deterministic decoding):

```
python src/run_llm_eval.py -extractor medgemma \
    -model google/medgemma-4b-it \
    -out ../paper02/outputs/llm_extractor_eval.json
```

Table 5: **Extractor comparison through the identical safety harness**, including an open medical LLM (MedGemma) measured against the same prespecified benchmark and schema-to-feature adapter as the rule-based references. End-to-end red-flag recall/specificity are the safety-critical numbers; the LLM never saw the benchmark cue lists. Generated by `code/src/run_llm_eval.py`.

| Extractor                | Macro F1 | Doc. exact | Urg.-chg. err. | E2E recall | E2E spec. | Run consist. |
|--------------------------|----------|------------|----------------|------------|-----------|--------------|
| Rule-based v1            | 0.72     | 0.59       | 0.27           | 56%        | 89%       | 1.00         |
| Rule-based v2 (improved) | 0.94     | 0.84       | 0.00           | 100%       | 100%      | 1.00         |
| MedGemma (open LLM)      | 0.75     | 0.57       | 0.11           | 78%        | 100%      | 1.00         |

Through this harness, MedGemma reached **78% end-to-end red-flag recall** (14/18; 95% Wilson CI 55–91%) at **100% specificity** (19/19; over-referral 0%), with field-level macro-F1 0.75 and a 10.8% urgency-changing error rate; greedy decoding made it perfectly reproducible (run-to-run consistency 1.00). It therefore sits *between* the two rule-based references—naïve 56%, MedGemma 78%, tuned 100%—on identical cases (Table 5). Its four missed urgent notes were the implicitly-phrased ones: recall was 88% on typical phrasing but fell to 71% (colloquial) and 67% (ambiguous), the same categories where the naïve rule extractor failed. Notably, although MedGemma *over-*asserted audiometric asymmetry at the field level (precision 0.13, liberally reading “asymmetry” from

narrative text), this did not produce a single over-referral: the conservative combined rule—single-sided tinnitus *with* asymmetry—absorbed the over-call, a direct validation of the combined-rule design. The plumbing itself is separately guaranteed by seven unit tests that exercise the full path (prompt construction, generation, code-fence/prose-tolerant JSON parsing, schema-to-feature mapping, safety layer, metrics) with a deterministic no-model stand-in.

This is the result that speaks to the circularity concern raised against author-curated benchmarks. MedGemma neither authored the benchmark nor saw its cue lists, yet was scored against the same prespecified target through the same adapter; its 78% is thus an *extractor-independent* measurement, not a re-scoring of the hand-tuned rules. Two things follow. First, the benchmark is *not saturated*: a capable general-purpose open medical LLM still misses four in eighteen urgent presentations, so a ceiling of 100% is specific to the tuned extractor, not the task. Second, the benchmark is *not gamed*: an independent model lands well below the tuned 100%, so that figure reflects benchmark-specific tuning and the honest deployment expectation for an off-the-shelf LLM is closer to 78%—below the level at which a safety layer could stand without mandatory clinician verification. The entire measurement is reproducible on a free GPU via the provided notebook.

## 4 Discussion

The central result is a clean separation of concerns. The deterministic layer is *perfectly reliable given correct features*: complete guideline coverage, zero over-referral, and full auditability, and it structurally guarantees that a low probabilistic score can never suppress referral. This is the property a safety-critical otology pipeline needs, and it is achieved without a model. But end-to-end safety is governed by the extraction step, and the benchmark *measures* how much: red-flag recall was 56% (naive extractor), 78% for an actual open medical LLM (MedGemma-4b-it, independent of the benchmark), and 100% (tuned extractor) on identical cases. A red flag the extractor never surfaces cannot be rescued by a downstream rule, and a resolved historical event can trigger a false alarm; the improvements that closed the rule-extractor gap were precisely historical-context suppression and better handling of colloquial and implicit phrasing—and MedGemma’s own misses fell in exactly those colloquial/ambiguous categories, so the bottleneck is intrinsic to extraction, not an artifact of one hand-written extractor.

The practical implications are specific. First, **extraction quality—not the rule layer—is the safety bottleneck**, and an open benchmark that quantifies it (as this one does, spanning 56%, 78%, and 100% recall across independent extractors) is a prerequisite for trustworthy deployment. Second, **a finite-set recall of 100% is a target, not a guarantee**: the tuned extractor is perfect *on these 71 curated cases*, but an off-the-shelf open medical LLM reaches only 78% on the same cases, and real notes are messier still. **Mandatory clinician verification of the extracted red-flag features is therefore load-bearing, not a formality**—the layer plus human-in-the-loop review, not the layer alone, is the safe configuration; a model at 78% recall would, unverified, let roughly one urgent case in five reach the “routine” path. The measured MedGemma result also makes the



bar concrete: an off-the-shelf open LLM, plugged into the same harness, does not yet match the tuned extractor and must improve on colloquial/ambiguous phrasing—the axes on which both it and the naive extractor failed—before it could be trusted without human review.

## 4.1 Limitations

This is a curated benchmark (71 cases), so confidence intervals are wide—MedGemma’s 78% recall carries a 55–91% Wilson interval—and subgroup counts are modest; the benchmark is designed to be extended, and the harness supports arbitrarily many cases. The MedGemma evaluation, though real, uses the 4B instruction-tuned checkpoint under zero-shot schema prompting on the *open* benchmark; larger MedGemma variants, few-shot prompting, and—above all—*real* clinical notes (which need IRB-approved, deidentified text) remain to be tested, and performance on messy real-world text may differ from these curated cases. The benchmark, metrics, and adapter were fixed before the model was run, so the MedGemma result is prespecified rather than post-hoc, and—because the model is independent of the benchmark’s authorship and cue lists—model-independent by construction. The tuned extractor’s perfect recall is on curated cases and must not be read as a deployment guarantee; the measured 78% for an off-the-shelf LLM is the more realistic signal. Cases are English-language; multilingual and real-note robustness (including real negation and temporal complexity) remain to be tested. Finally, a finite-set recall of 1.00 at the rule-coverage level is a target, not a guarantee of zero deployment misses; the layer supplements, never replaces, clinical judgment.

## 5 Conclusion

A deterministic, guideline-derived safety layer can guarantee that a probabilistic model never suppresses referral for a time-critical otologic emergency, and—given correctly-extracted features—misses no guideline red flag while over-referring none. But end-to-end safety is only as good as the extraction that feeds it: on free text, red-flag recall was 56%, 78%, and 100% across a naive extractor, a real open medical LLM (MedGemma-4b-it), and a tuned extractor on identical cases, so extraction quality is the safety bottleneck and clinician verification is essential. By releasing the rules, the open benchmark, reproducible extractors, a schema-to-feature adapter, the metrics, and an executable, unit-validated open medical-LLM evaluation harness, we turn a safety *claim* into a prespecified, reproducible, and model-independent *evaluation*—and use it to measure an actual open medical LLM (MedGemma at 78% recall), an extractor-independent result that resolves the circularity concern of a self-authored benchmark and is reproducible on a free GPU.

## Declarations

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read and approved the manuscript. **Data and Code Availability:** The benchmark, deterministic rules, rule-based extractors, the open medical-LLM (MedGemma) extractor path, the unified evaluation harness, its end-to-end tests, and the generated result tables are openly available in the STARS repository (`code/src/redflag_benchmark.py`, `safety.py`, `llm_extract.py`, `llm_medgemma.py`, `run_redflag_eval.py`, `run_extraction_eval.py`, `run_llm_eval.py`, `test_llm_medgemma.py`), together with a ready-to-run free-GPU notebook (`code/notebooks/medgemma_eval_colab.ipynb`) that reproduces the MedGemma row on a free Colab/Kaggle T4. No patient data are used or redistributed. **Ethics:** No human-subjects data; all cases are expert-authored or synthetic. A prospective clinical evaluation will proceed only under IRB approval. **AI-Use Disclosure:** Generative AI tools assisted in drafting and code scaffolding; the author reviewed and is responsible for all content. In any clinical use, open medical LLMs are confined to clinician-verified extraction and never make autonomous decisions. **Conflicts of Interest:** None declared. **Funding:** None.

## References

- [1] Chandrasekhar SS, Tsai Do BS, Schwartz SR, et al. Clinical practice guideline: sudden hearing loss (update). *Otolaryngol Head Neck Surg.* 2019;161(1\_suppl):S1–S45. <https://doi.org/10.1177/0194599819859885>
- [2] Tunkel DE, Bauer CA, Sun GH, et al. Clinical practice guideline: tinnitus. *Otolaryngol Head Neck Surg.* 2014;151(2\_suppl):S1–S40. <https://doi.org/10.1177/0194599814545325>
- [3] Agrawal M, Hegselmann S, Lang H, Kim Y, Sontag D. Large language models are few-shot clinical information extractors. In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics; 2022:1998–2022.
- [4] Uzuner Ö, South BR, Shen S, DuVall SL. 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text. *J Am Med Inform Assoc.* 2011;18(5):552–556.
- [5] Singhal K, Azizi S, Tu T, et al. Large language models encode clinical knowledge. *Nature.* 2023;620(7972):172–180.
- [6] Google Research. MedGemma: open models for health AI development. 2025. Accessed August 8, 2026. <https://research.google/blog/medgemma-our-most-capable-open-models-for-health-ai-development/>
- [7] Google Health AI Developer Foundations. MedGemma. 2026. Accessed August 8, 2026. <https://developers.google.com/health-ai-developer-foundations/medgemma>
- [8] Wong A, Otles E, Donnelly JP, et al. External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. *JAMA Intern Med.* 2021;181(8):1065–1070.
- [9] Finlayson SG, Subbaswamy A, Singh K, et al. The clinician and dataset shift in artificial intelligence. *N Engl J Med.* 2021;385(3):283–286.

- [10] Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*. 2024;385:e078378. <https://doi.org/10.1136/bmj-2023-078378>